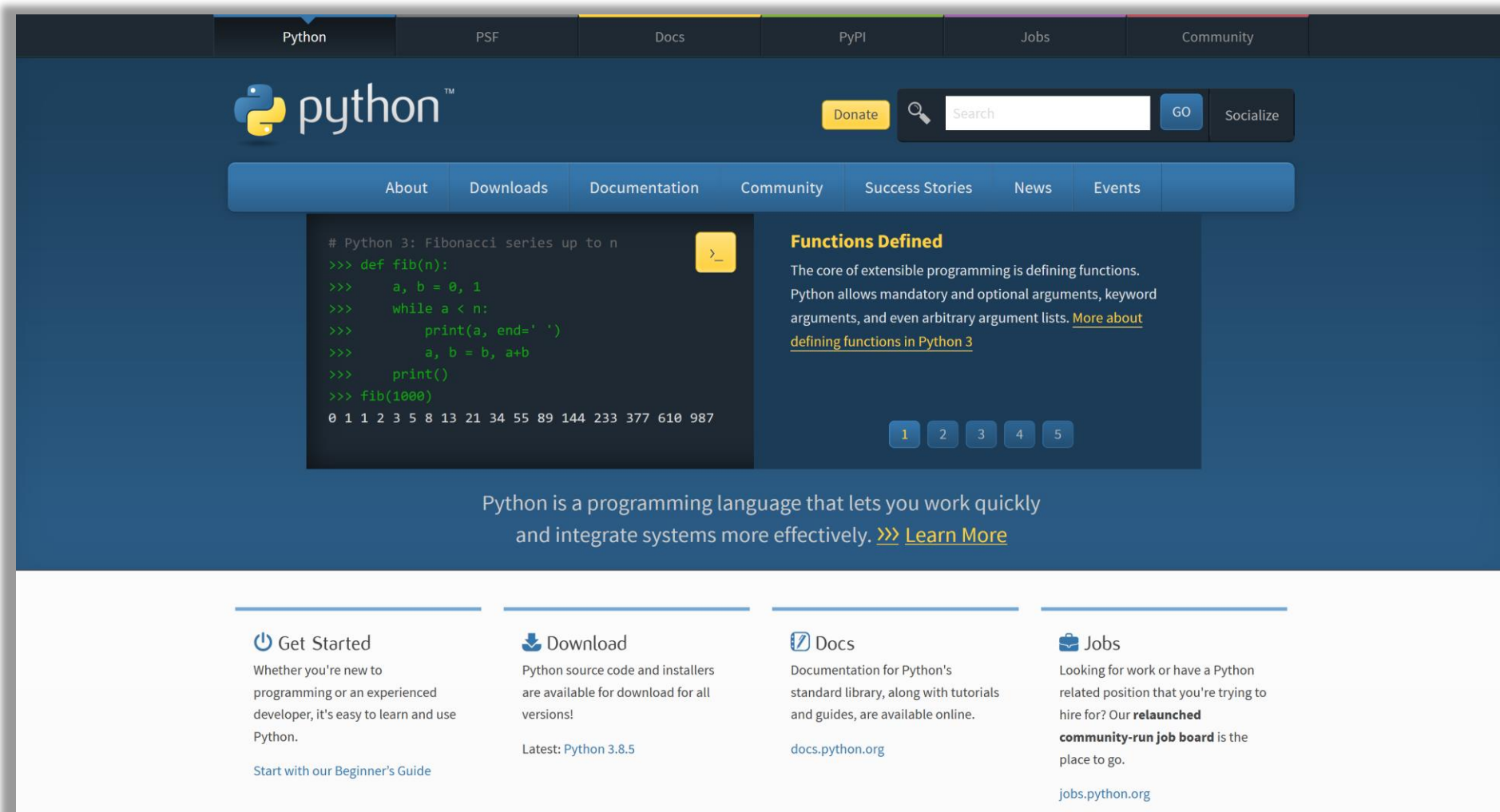


Familiarizing with VSCode and Python

<https://www.python.org/>



The screenshot shows the Python.org homepage with a dark blue header and a lighter blue main content area. The header includes navigation links for Python, PSF, Docs, PyPI, Jobs, and Community. The main content area features the Python logo, a search bar, and a navigation bar with links for About, Downloads, Documentation, Community, Success Stories, News, and Events. The central content area is divided into two columns: the left column displays a code snippet for a Fibonacci series, and the right column contains the 'Functions Defined' section. Below this, a banner states 'Python is a programming language that lets you work quickly and integrate systems more effectively.' with a 'Learn More' link. The footer contains four sections: 'Get Started', 'Download', 'Docs', and 'Jobs', each with a brief description and a link to further resources.

Python

PSF

Docs

PyPI

Jobs

Community

python™

Donate

Search

GO

Socialize

About

Downloads

Documentation

Community

Success Stories

News

Events

```
# Python 3: Fibonacci series up to n
>>> def fib(n):
>>>     a, b = 0, 1
>>>     while a < n:
>>>         print(a, end=' ')
>>>         a, b = b, a+b
>>>     print()
>>> fib(1000)
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987
```

Functions Defined

The core of extensible programming is defining functions. Python allows mandatory and optional arguments, keyword arguments, and even arbitrary argument lists. [More about defining functions in Python 3](#)

1 2 3 4 5

Python is a programming language that lets you work quickly and integrate systems more effectively. [>>> Learn More](#)

Get Started

Whether you're new to programming or an experienced developer, it's easy to learn and use Python.

[Start with our Beginner's Guide](#)

Download

Python source code and installers are available for download for all versions!

Latest: Python 3.8.5

Docs

Documentation for Python's standard library, along with tutorials and guides, are available online.

docs.python.org

Jobs

Looking for work or have a Python related position that you're trying to hire for? Our **relaunched community-run job board** is the place to go.

jobs.python.org

<https://docs.python.org/>

Python » English » 3.8.5 » Documentation »

Quick search | [modules](#) | [index](#)

Download

Download these documents

Docs by version

- [Python 3.10 \(in development\)](#)
- [Python 3.9 \(pre-release\)](#)
- [Python 3.8 \(stable\)](#)
- [Python 3.7 \(security-fixes\)](#)
- [Python 3.6 \(security-fixes\)](#)
- [Python 3.5 \(security-fixes\)](#)
- [Python 2.7 \(EOL\)](#)
- [All versions](#)

Other resources

- [PEP Index](#)
- [Beginner's Guide](#)
- [Book List](#)
- [Audio/Visual Talks](#)
- [Python Developer's Guide](#)

Python 3.8.5 documentation

Welcome! This is the documentation for Python 3.8.5.

Parts of the documentation:

- [What's new in Python 3.8?](#)
or all "What's new" documents since 2.0
- [Tutorial](#)
start here
- [Library Reference](#)
keep this under your pillow
- [Language Reference](#)
describes syntax and language elements
- [Python Setup and Usage](#)
how to use Python on different platforms
- [Python HOWTOs](#)
in-depth documents on specific topics
- [Installing Python Modules](#)
installing from the Python Package Index & other sources
- [Distributing Python Modules](#)
publishing modules for installation by others
- [Extending and Embedding](#)
tutorial for C/C++ programmers
- [Python/C API](#)
reference for C/C++ programmers
- [FAQs](#)
frequently asked questions (with answers!)

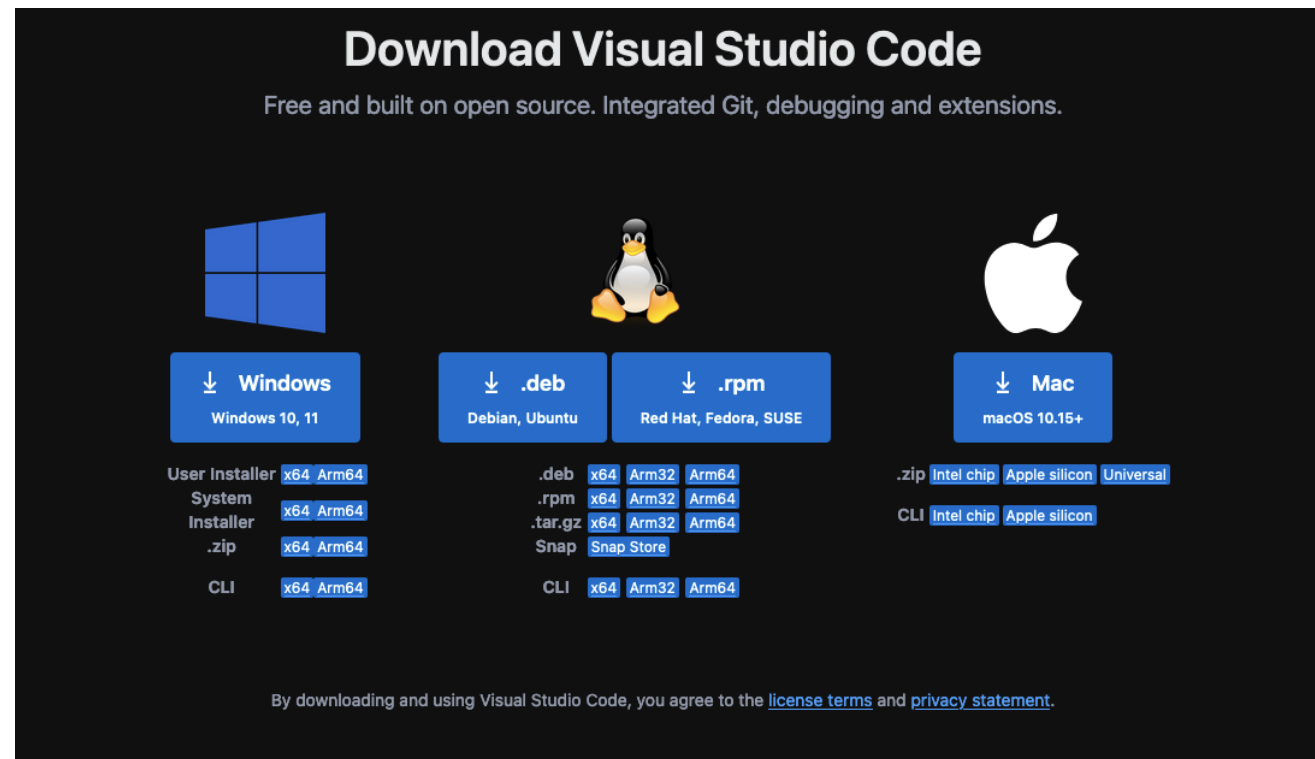
Indices and tables:

- [Global Module Index](#)
quick access to all modules
- [General Index](#)
all functions, classes, terms
- [Glossary](#)
the most important terms explained
- [Search page](#)
search this documentation
- [Complete Table of Contents](#)
lists all sections and subsections

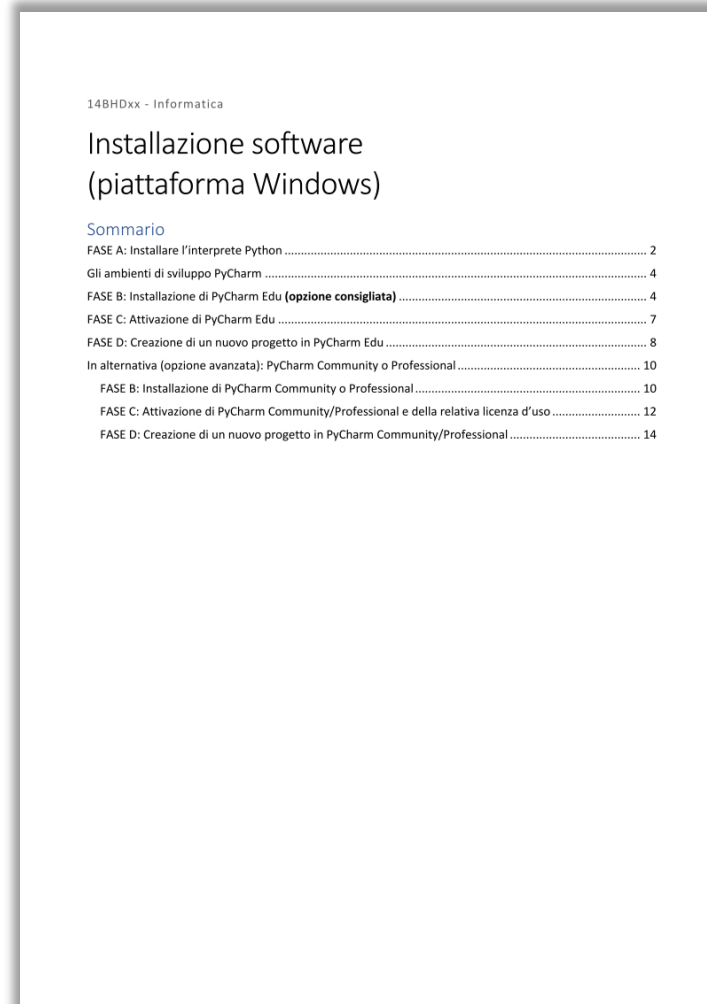
Meta information:

IDE

- Install Visual Studio Code and the Python Extension
- <https://code.visualstudio.com/download>



Installation instructions (in the Course Page)



148HDxx - Informatica

Installazione software (piattaforma Windows)

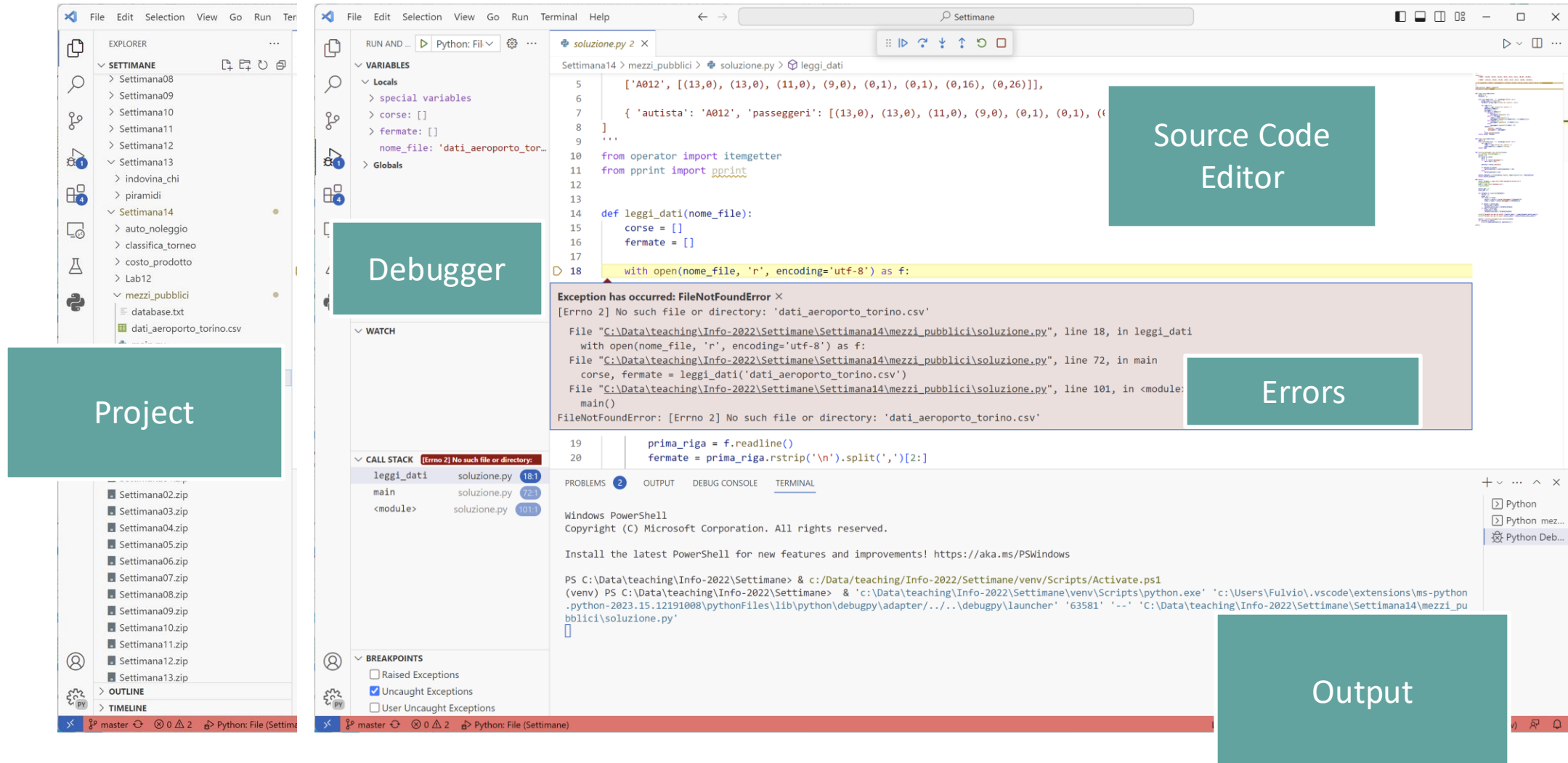
[Sommar](#)

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IDE components

- Most IDEs (not only VSCode) usually include:
- **The source code editor**: a text editor with additional functionality
 - Listing line numbers of code
 - Syntax highlighting and coloring (comments, text...)
 - Auto-indent source code
 - Highlight syntax errors
 - Auto-completion
- **Output window**
 - The output generated by the program
- **Debugger**
 - Tools to help you search and correct logic errors in the program

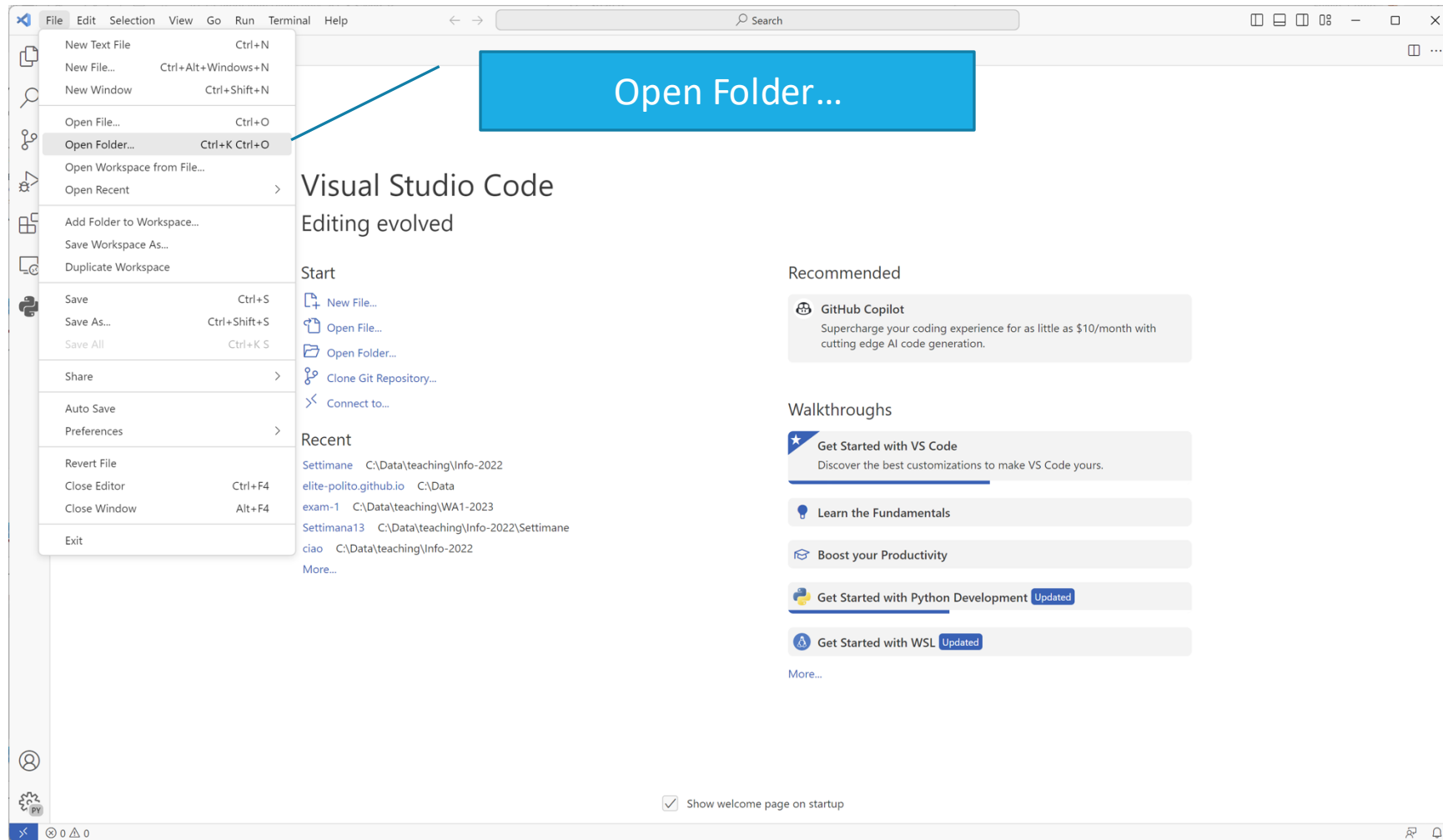
The Visual Studio Code IDE



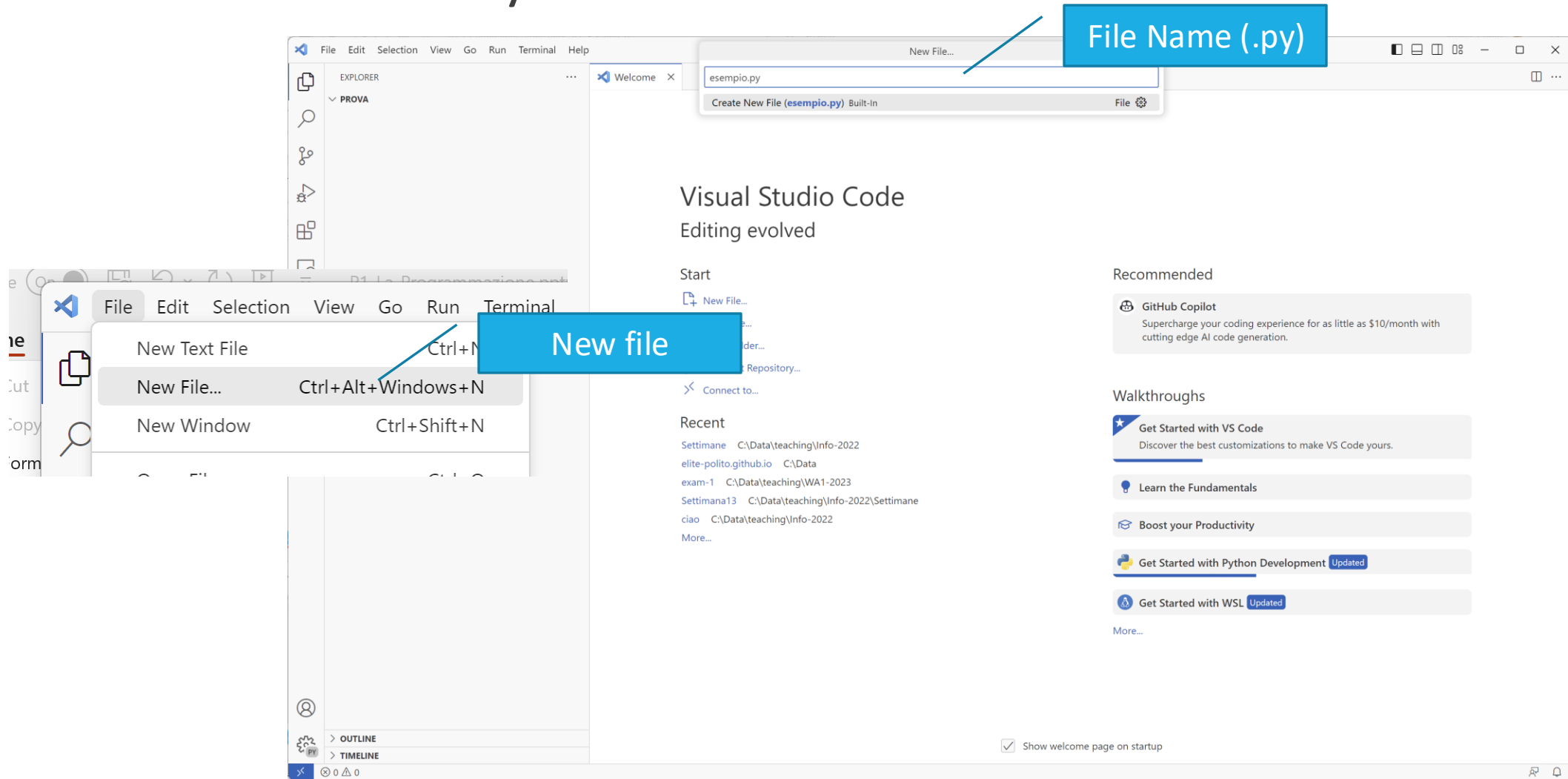
Project vs Program vs File

- A single **Program** may be very large, and will be composed of many different **files**
- IDEs allow us to group a set of related files into a “**Project**”
 - In VSCode, a project corresponds to a folder.
- Every time we want to create a new **program**, we must
 - Create a new **Project (folder)**
 - Create one (or more) Python **Files** inside the **Project**

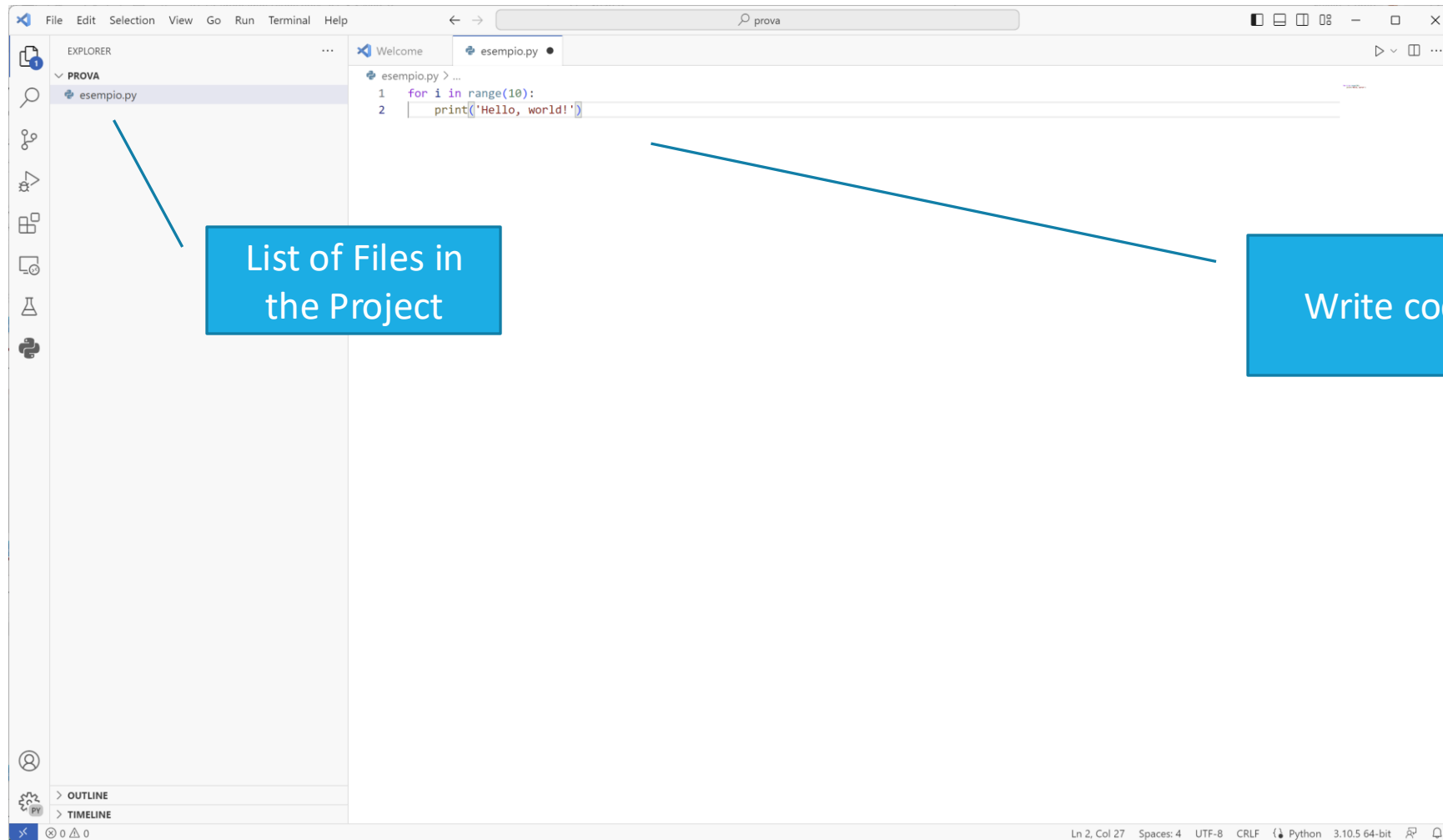
Create a New Project



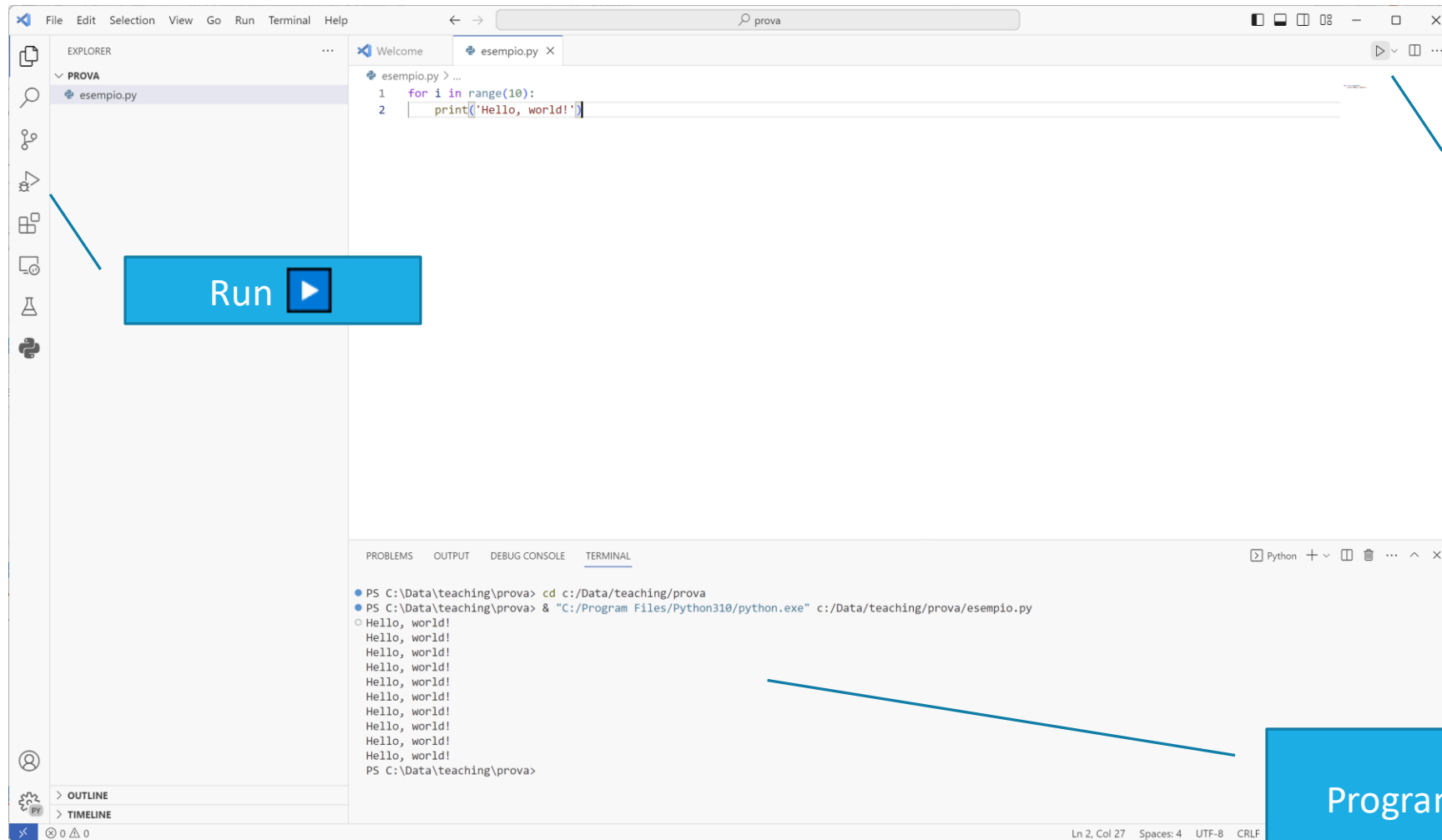
Create a Python File



Write a Python Program



Execute the Program



Organize your work

- Your 'source code' is stored in .py files
- Create a folder for this course
- Create one **project folder** per program inside the course folder
 - A program can consist of several .py files
- Backup your files regularly
 - To a USB flash drive (or more than one)
 - To a network drive (cloud service) or external hard drive
 - Do it. Really.

Python Interactive Mode

- The Python interpreter can load a file and execute all the instructions contained in it (and other files from the same project)
- Alternatively: in **interactive mode**, you can run instructions one at a time
 - It allows quick 'test programs' to be written
 - Try and experiment instructions

Using Interactive Mode in VSCode

Press F1 to open the VSCode Command Window

Type REPL and select Python: Start Terminal REPL

REPL = Read, Evaluate, Print Loop

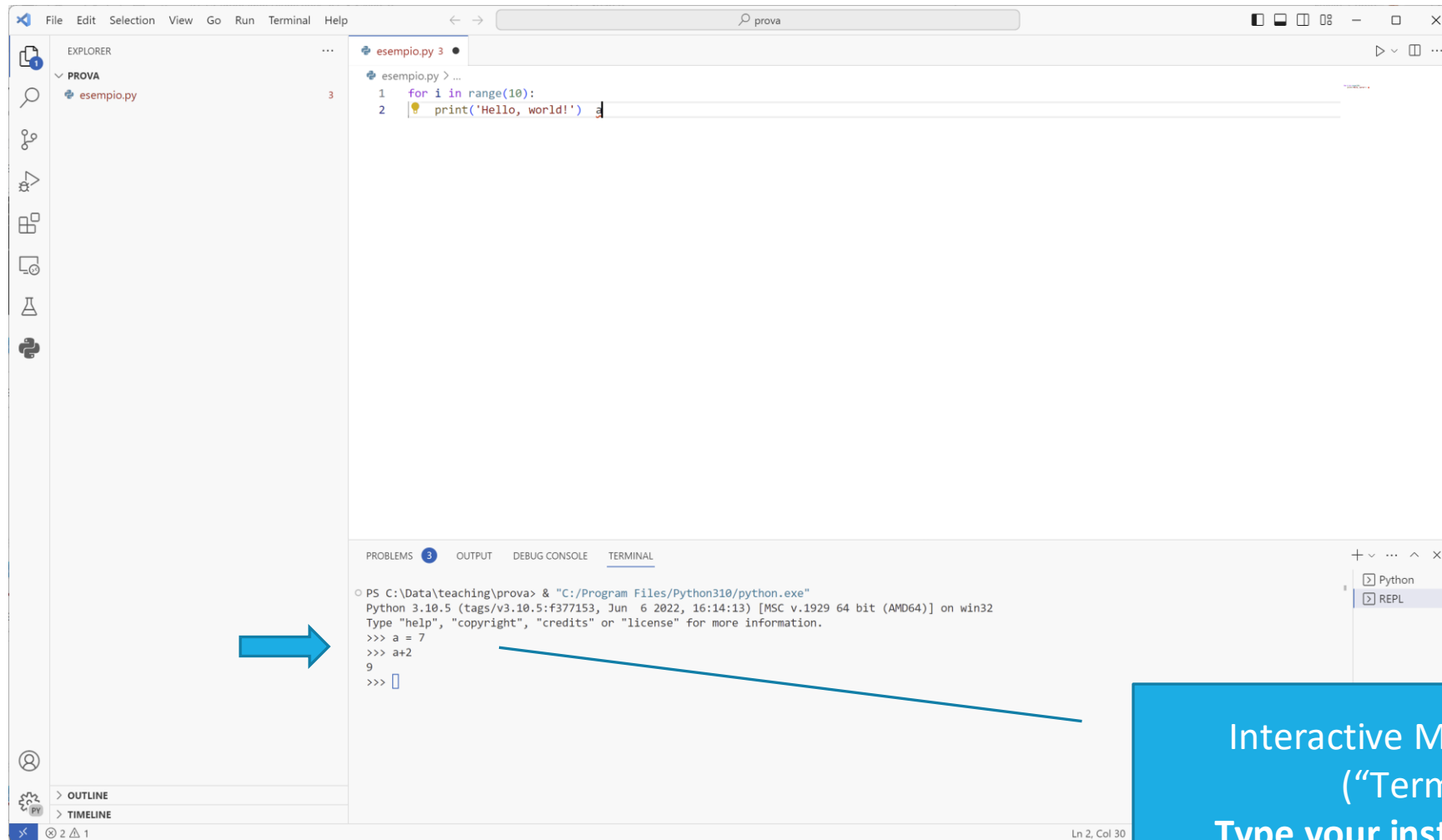
```
Python: Start REPL
Replace
Replace with Next Value
Replace with Previous Value
Search: Replace in Files

Python REPL
> rep|

Unable to initialize device PRN
PS C:\Data\teaching\prova> print('Hello, world!')
Unable to initialize device PRN
PS C:\Data\teaching\prova>
>> a=7
a=7 : The term 'a=7' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:2 char:1
+ a=7
+ ~~~
+ CategoryInfo          : ObjectNotFound: (a=7:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Data\teaching\prova>
PS C:\Data\teaching\prova> |
```

Using Interactive Mode in VSCode



Interactive Mode Window
("Terminal")
Type your instructions here

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